

Hik-Partner Pro & Hik-Connect for Teams Network SDK （Android）

Developer Guide

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Contents

Chapter 1 Overview 1

1.1 Introduction 1

1.2 Development Environment 1

1.3 Dependency Libraries 1

1.4 Update History 1

Chapter 2 Typical Applications 3

2.1 Initialization 3

2.2 Start Live View 3

2.3 Start Two-Way Audio 5

2.4 Start Playback 6

Chapter 3 API Reference 8

3.1 Basic APIs 8

3.1.1 initWithAreaDomain 8

3.1.2 getLastError 9

3.1.3 getSDKVersion 9

3.2 Live View Related APIs 10

3.2.1 createPreviewPlayer 10

3.2.2 startRealPlay 10

3.2.3 openPlaySound 11

3.2.4 closePlaySound 11

3.2.5 stopPlay 12

3.2.6 changeStreamType 12

3.2.7 capturePicture 13

3.2.8 startLocalRecord 13

3.2.9 stopLocalRecord 14

3.2.10 electricZoom 14

3.2.11 closeEletricZoom 15
3.3 Two-Way Audio Related APIs 15
3.3.1 createIntercom 15
3.3.2 startVoiceTalk 16
3.3.3 stopVoiceTalk 17
3.4 Playback Related APIs 18
3.4.1 createPlayBackPlayer 18
3.4.2 searchRecordFile 18
3.4.3 startPlayback 19
3.4.4 pausePlayback 20
3.4.5 resumePlayback 20
3.4.6 openPlaySound 20
3.4.7 closePlaySound 21
3.4.8 stopPlay 21
3.4.9 capturePicture 22
3.4.10 startLocalRecord 22
3.4.11 stopLocalRecord 23
3.4.12 setPlaybackSpeed 23
3.4.13 electricZoom 24
3.4.14 closeEletricZoom 24
3.4.15 getCapacitySupport 25
3.4.16 startDownloadFile 25
3.4.17 stopDownloadFile 26
Appendix A. Callback Function 28
A.1 InitCallback 28
A.2 PreviewCallback 28
A.3 PlaybackCallback 29
A.4 IntercomCallback 31

A.5 DownloadFileResultCallback 31
Appendix B. Data Structure 33
B.1 HPNetError 33
B.2 HPNetPlayerParam 33
B.3 HPNetIntercomParam 33
B.4 HPNetRecordFile 33
B.5 HPNetSDKException 34
Appendix C. HPNetSDK Status Code 35
Appendix D. HPNetSDK Error Code 36

Chapter 1 Overview

1.1 Introduction

Hik-Partner Pro Network SDK (hereinafter referred to as "HPNetSDK") is a software development kit that provides multiple APIs to implement some basic functions (such as initialization, live view, playback, picture capturing, digital zoom, and so on).

This guide mainly introduces the HPNetSDK based on Koltin programming language, the API calling flow of typical applications, and sample codes.

Note

HPNetSDK supports the real device testing only, it does not support the simulator testing.

1.2 Development Environment

The development environment of HPNetSDK based on Koltin programming language is shown in the table below.

Item	Required
Operating System	Android 5.0 or above

1.3 Dependency Libraries

Import the following third-party dependency libraries.

com.squareup.okhttp3:okhttp

com.google.code.gson:gson

com.squareup.okio:okio

com.squareup.retrofit2:retrofit

com.squareup.retrofit2:converter-gson

com.squareup.retrofit2:adapter-rxjava2

com.squareup.okhttp3:logging-interceptor

io.reactivex.rxjava2:rxjava

io.reactivex.rxjava2:rxandroid

1.4 Update History

Summary of Changes in Version 3.0_Nov., 2024

1. Added a typical application ***Start Two-Way Audio*** and 3 related APIs:
   - Create two-way audio instance: ***createIntercom*** ;
   - Start two-way audio: ***startVoiceTalk*** ;
   - Stop two-way audio: ***stopVoiceTalk*** .
2. Corrected ***HPNetSDK Status Code*** , and added 3 error codes (HPNETSDK1-100017, HPNETSDK1-100018, and HPNETSDK4-510000) in ***HPNetSDK Error Code*** .

Summary of Changes in Version 2.1.0_Feb., 2023

1. Added one API for starting downloading files: ***startDownloadFile*** .
2. Added one API for stopping downloading files: ***stopDownloadFile*** .
3. Added four error codes: Encapsulation failed; Starting downloading failed; Stopping downloading failed; Unknown error. For details, refer to ***HPNetSDK Error Code*** .

Summary of Changes in Version 1.0.0_Aug., 2021

New document.

Chapter 2 Typical Applications

2.1 Initialization

The diagram below shows the workflow of SDK initialization.

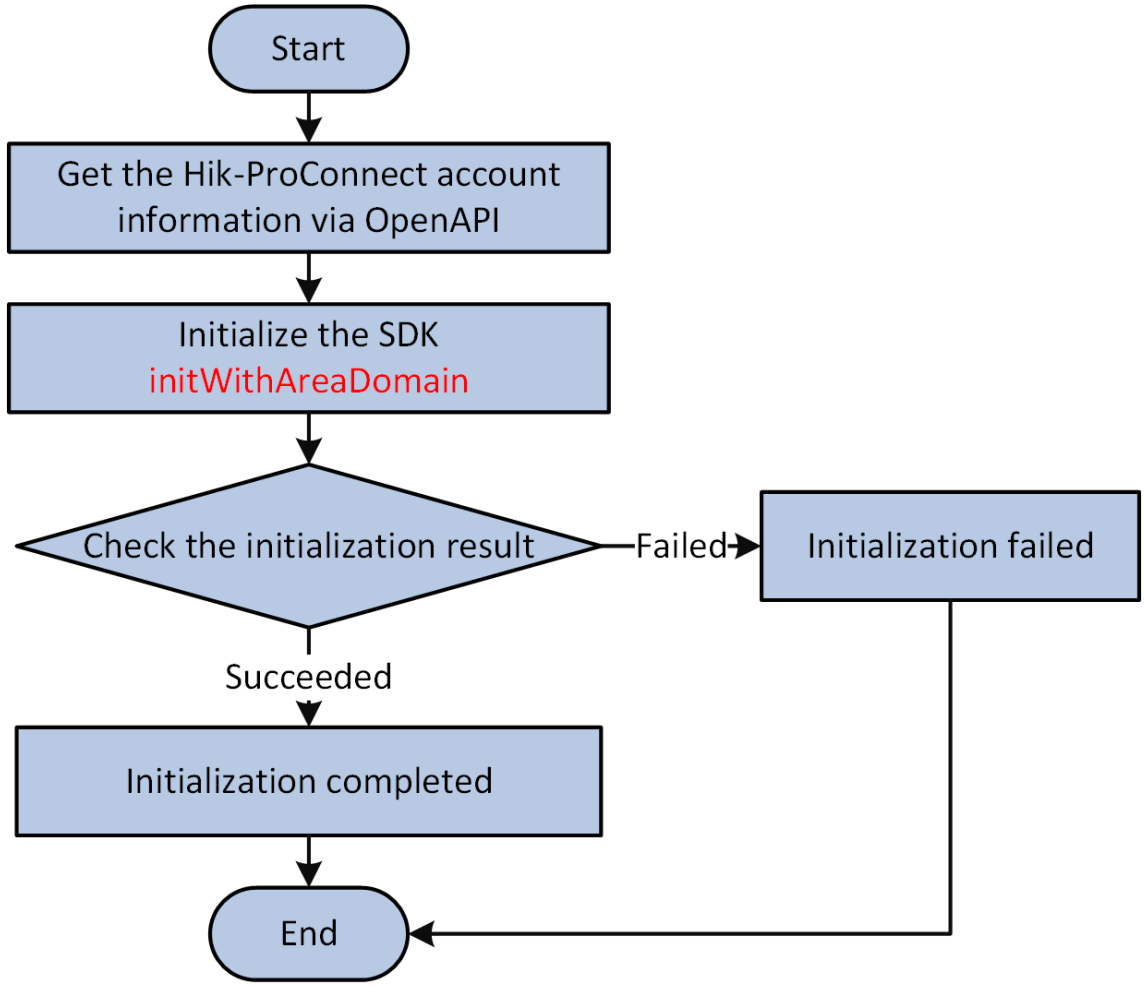


Figure 2-1 API Calling Flow of SDK Initialization

Related API(s)

Initialize the SDK: ***initWithAreaDomain***

2.2 Start Live View

Before You Start

Make sure you have initialized the SDK, including the development environment and resources.

Steps

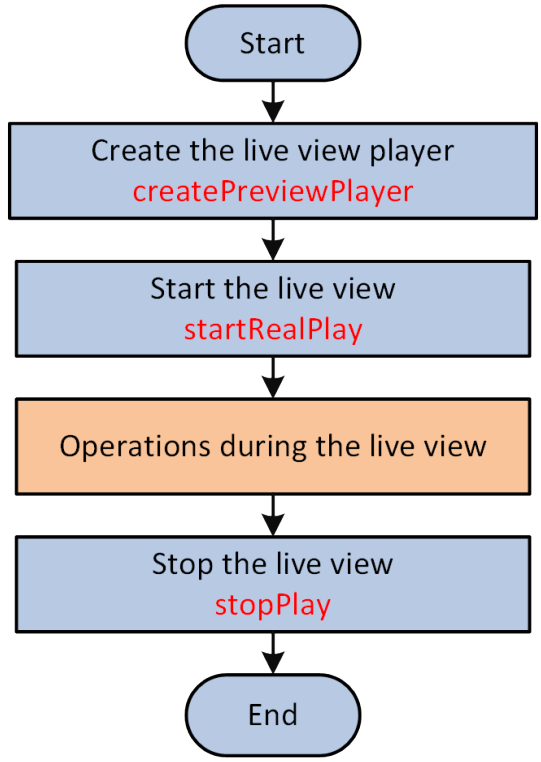


Figure 2-2 API Calling Flow of Starting Live View

1. Call ***createPreviewPlayer*** to create the live view player.
2. Call ***startRealPlay*** to start the live view.
3. **Optional:** Perform the following operations during the live view.

Function	API
Turn on the audio	***openPlaySound***
Turn off the audio	***closePlaySound***
Set the video definition	***changeStreamType***
Capture pictures	***capturePicture***
Start recording	***startLocalRecord***
Stop recording	***stopLocalRecord***
Enable the digital zoom	***electricZoom***
Disable the digital zoom	***closeEletricZoom***
Start the two-way audio	***startVoiceTalk***
Stop the two-way audio	***stopVoiceTalk***

4. Call ***stopPlay*** to stop the live view.

2.3 Start Two-Way Audio

Before You Start

Make sure you have initialized the SDK, including the development environment and resources.

Steps

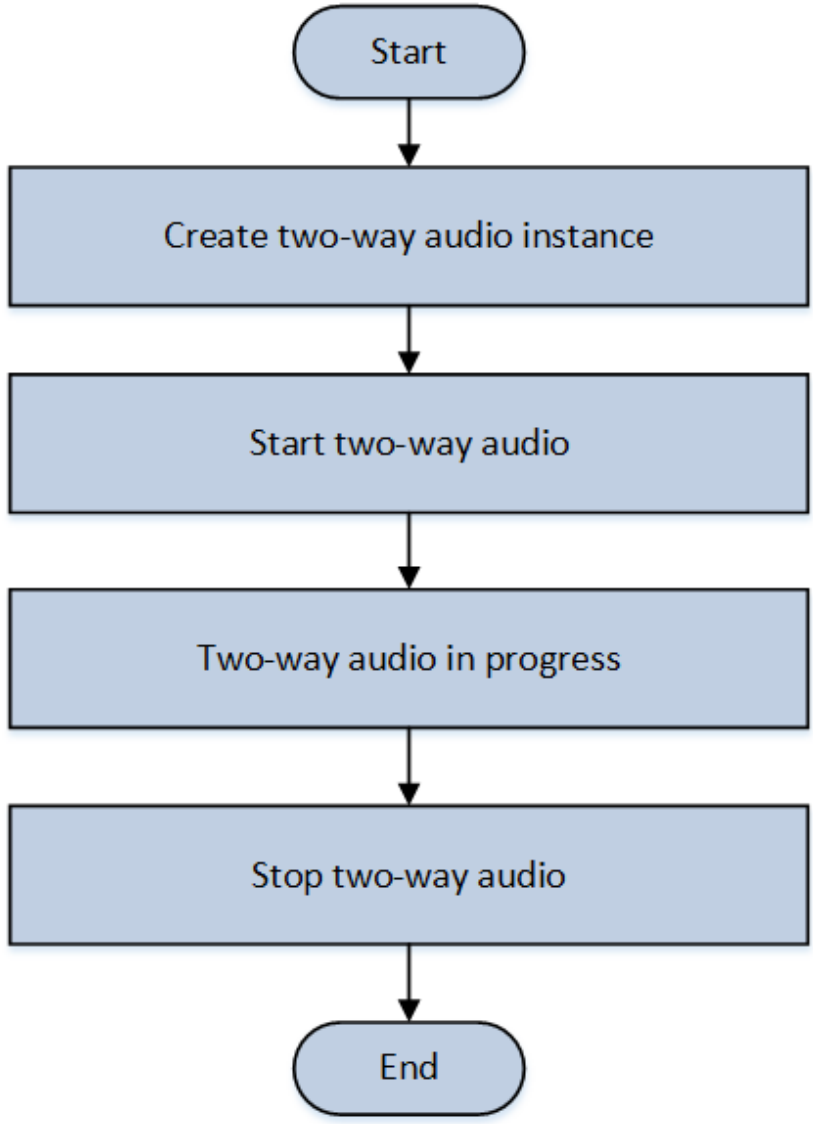


Figure 2-3 API Calling Flow of Starting Two-Way Audio

1. Call ***createPreviewPlayer*** to create the two-way audio example.
2. Call ***startVoiceTalk*** to start two-way audio.
3. Call ***stopVoiceTalk*** to stop two-way audio.

2.4 Start Playback

You can search for the camera's video files in specified time period for playback, and perform further operations, such as pausing or resuming playback, controlling playback speed, and controlling the audio during playback.

Before You Start

Make sure you have initialized the SDK, including the development environment and resources.

Steps

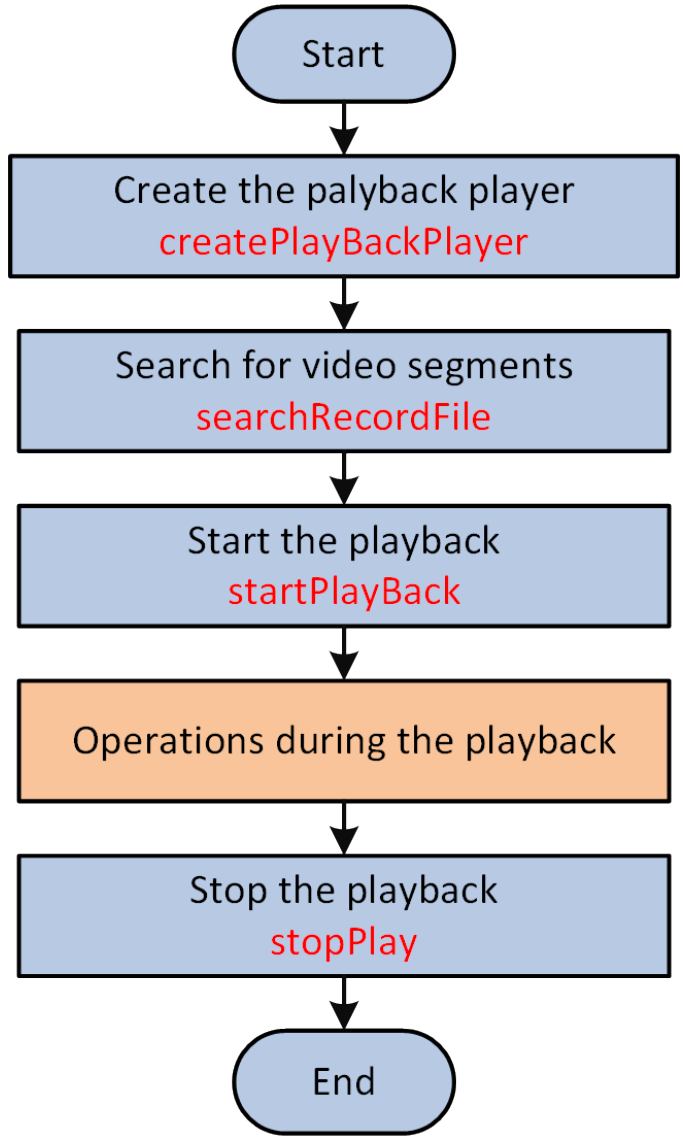


Figure 2-4 API Calling Flow of Starting Playback

1. Call ***createPlayBackPlayer*** to create the playback player.

2. Call ***searchRecordFile*** to search for video segments.
3. Call ***startPlayback*** to start the playback.
4. **Optional:** Perform following operations during the playback.

Function	API
Pause the playback	***pausePlayback***
Resume the playback	***resumePlayback***
Turn on the audio	***openPlaySound***
Turn off the audio	***closePlaySound***
Capture pictures	***capturePicture***
Start recording	***startLocalRecord***
Stop recording	***stopLocalRecord***
Set the playback speed	***setPlaybackSpeed***
Enable the digital zoom	***electricZoom***
Disable the digital zoom	***closeEletricZoom***
Get the capability set	***getCapacitySupport***

5. Call ***stopPlay*** to stop the playback.

Chapter 3 API Reference

3.1 Basic APIs

3.1.1 initWithAreaDomain

Initialize the SDK.

API Definition

```
+fun initWithAreaDomain(
   application: Application,
   areaDomain: String,
   accessToken: String,
   appKey: String,
   callback: InitCallback
)
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
areaDomain	String	Req.	The domain name of the Hik-Partner Pro OpenAPI. The **accessToken** is valid within this domain only.
accessToken	String	Req.	API access token
appkey	String	Req.	Developer account (AK)
application	Application	Req.	Android application instance

Return Value

Return the initialization result asynchronously in the callback function ***InitCallback*** .

Example

Sample Code of Initialization

```
HPNetSDK.initWithAreaDomain(
   DemoApplication.instance,
   "https://ieuapi.hik-proconnect.com",
   "jmna8qnqg8d3dgnzs87m4v2dme3l",
   "dfgi8qnqg8d3dlief87m4v2diemc",
   object : InitCallback {
      /**
       * Initialized.
```

```
        */
        override fun onInitSuccess() {
        }
        /**
         * Initialization failed.
         */
        override fun onInitFail(error: HPNetError) {
        }
    }
)
```

3.1.2 getLastError

Get the error code.

API Definition

```
+ fun getLastError() :HPNetError?
```

Return Value

Return the error code instance, see ***HPNetError*** for details.

Example

Sample Code of Getting the Error Code

```
val error = ErrorManager.getLastError()
```

3.1.3 getSDKVersion

Get the SDK version information.

API Definition

```
+ fun getSDKVersion():String
```

Return Value

Return the SDK version information, e.g., V1.0.0 Build20210524.

Example

Sample Code of Getting the SDK Version Information

```
val sdkVersion = HPNetSDK.getSDKVersion()
```

3.2 Live View Related APIs

3.2.1 createPreviewPlayer

Create the live view player.

API Definition

```
+ fun createPreviewPlayer(param: HPNetPlayerParam, callback: PreviewCallback): HPNetPreviewPlayer?
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
param	HPNetPlayerParam	Req.	Live view player parameters. Refer to ***HPNetPlayerParam*** for details.
callback	PreviewCallback	Req.	The callback instance. Refer to ***PreviewCallback*** for details of the callback method.

Return Value

Return the live view player instance, which is used for later operations related to the live view.

Example

Sample Code of Creating the Live View Player

```
val param = HPNetPlayerParam("123456789",
        1,
        surfaceView)
val mPlayer = HPNetSDK.createPreviewPlayer(param, this)
```

3.2.2 startRealPlay

Start the live view.

API Definition

```
- fun startRealPlay(@Type.StreamType streamType: Int, safeKey:String?)
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
streamType	Int	Req.	The stream type: Type.STREAM_LOW (fluent), Type.STREAM_MID (balanced), Type.STREAM_HIGH (HD).
safeKey	String	Req.	The verification code of video encryption.

Return Value

No value is returned. The live view result will be notified via the callback ***PreviewCallback*** .

Example

Sample Code of Starting the Live View

```
mPlayer.startRealPlay(Type.STREAM_LOW,secretKey);
```

3.2.3 openPlaySound

Turn on the audio during the live view or playback.

API Definition

```
- fun openPlaySound() :Boolean
```

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example

Sample Code of Turning On the Audio

```
val result = mPlayer.openPlaySound()
```

3.2.4 closePlaySound

Turn off the audio during the live view or playback.

API Definition

```
- fun closePlaySound():Boolean
```

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example
Sample Code of Turning Off the Audio

```
val result = mPlayer.closePlaySound()
```

3.2.5 stopPlay

Stop the live view or playback.

API Definition

```
- fun stopPlay() :Boolean
```

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example
Sample Code of Stopping the Live View or Playback

```
val result = mPlayer.stopPlay()
```

3.2.6 changeStreamType

Set the video definition.

API Definition

```
- fun changeStreamType(streamType: Int)
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
streamType	Int	Req.	The stream type: Type.STREAM_LOW (fluent), Type.STREAM_MID (balanced), Type.STREAM_HIGH (HD).

Return Value

No value is returned. The definition configuration result will be notified via the callback ***PreviewCallback*** .

Example

Sample Code of Setting the Video Definition

```
mPlayer.changeStreamType(Type.STREAM_LOW);
```

3.2.7 capturePicture

Capture pictures during the live view or playback.

API Definition

```
- fun capturePicture(filePath: String): Boolean
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
filePath	String	Req.	Path for saving files

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example

Sample Code of Capturing Pictures

```
val result = mPalyer.capturePicture(filePath)
```

3.2.8 startLocalRecord

Start recording during the live view or playback.

API Definition

```
- fun startLocalRecord(): Boolean
```

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example

Sample Code of Starting Recording

```
val result = mPlayer.startLocalRecord()
```

Remarks

- For the video storage path configuration of live view, refer to ***PreviewCallback*** .
- For the video storage path configuration of playback, refer to ***PlaybackCallback*** .

3.2.9 stopLocalRecord

Stop recording during the live view or playback.

API Definition

```
- fun stopLocalRecord(): Boolean
```

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example

Sample Code of Stopping Recording

```
val result = mPlayer.stopLocalRecord()
```

3.2.10 electricZoom

Zoom in or out during the live view and playback.

API Definition

```
- fun electricZoom(originalRect: RectF, targetRect: RectF): Boolean
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
originalRect	RectF	Req.	Original area
targetRect	RectF	Req.	Target area

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example

Sample Code of Digital Zoom

```
val result = mPlayer.electricZoom(originalRect, targetRect)
```

3.2.11 closeEletricZoom

Disable the digital zoom during the live view or playback.

API Definition

```
- fun closeElectricZoom(): Boolean
```

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example

Sample Code of Disabling Digital Zoom

```
val result = mPlayer.closeElectricZoom()
```

3.3 Two-Way Audio Related APIs

3.3.1 createIntercom

Create two-way audio instance.

API Definition

```
+ fun createIntercom(
      param: HPNetIntercomParam,
      callback: IntercomCallback
  )
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
param	HPNetIntercomParam	Req.	Player parameters. Refer to

Parameter	Data Type	Req. or Opt.	Description
			HPNetIntercomParam for details.
callback	IntercomCallback	Req.	The callback instance. Refer to ***IntercomCallback*** for details.

Return Value

The two-way audio instance is returned for subsequent operations on two-way audio.

Example

Sample Code of Starting the Two-Way Audio

```
val param = HPNetIntercomParam(devSerial = "123456789", channelNo = 1)
val mIntercom = HPNetSDK.createIntercom(param = param, callback = object : IntercomCallback {
    override fun didReceiveMessage(intercom: IIntercom, msgCode: Int) {
        when(msgCode) {
            MessageCode.VOICE_TALK_START -> {
                // TODO:  start
            }
            MessageCode.VOICE_TALK_FINISH -> {
                // TODO:  finish
            }
            MessageCode.VOICE_TALK_AUTO_STOP -> {
                // TODO:  exceed time limit
            }
            MessageCode.VOICE_TALK_TOKEN_CHANGED -> {
                // TODO:  token changed
            }
        }
    }

    override fun didReceiveError(intercom: IIntercom, error: HPNetError) {
        //Log.e(TAG, "code: ${error.code}, desc: ${error.message}")
    }
})
```

3.3.2 startVoiceTalk

Start the two-way audio.

API Definition

```
+ fun startVoiceTalk(
        isDeviceTalk: Boolean = true,
```

```
    safeKey: String? = null
)
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
isDeviceTalk	Boolean	Opt.	Whether the two-way audio is via device: true (by default, two-way audio via device), false (two-way audio via channel).
safeKey	String?	Opt.	Streaming secret key (if the device is encrypted), null by default.

Return Value

No value is returned. The result of starting two-way audio will be notified via the callback ***IntercomCallback*** .

3.3.3 stopVoiceTalk

Stop the two-way audio.

API Definition

```
-(void)stopVoiceTalk;
```

Return Value

No value is returned. The result of stopping two-way audio will be notified via the callback ***IntercomCallback*** .

Example

Sample Code of Stopping the Two-Way Audio

```
mIntercom.stopVoiceTalk()
```

3.4 Playback Related APIs

3.4.1 createPlayBackPlayer

Create the playback player.

API Definition

```
+ fun createPreviewPlayer(param: HPNetPlayerParam, callback: PreviewCallback): HPNetPreviewPlayer?
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
param	HPNetPlayerParam	Req.	Player parameters. Refer to ***HPNetPlayerParam*** for details.
callback	PlaybackCallback	Req.	The callback instance. Refer to ***PlaybackCallback*** for details.

Return Value

Return the playback player instance for later operations related to the playback.

Example

Sample Code of Creating the Playback Player

```
val param = HPNetPlayerParam("123456789",
            1,
            mSurfaceView)
val mPlayer = HPNetSDK.createPlaybackPlayer(param, this)
```

3.4.2 searchRecordFile

Search video segments.

API Definition

```
@WorkerThread
@Throws(HPNetSDKException::class)
- fun searchRecordFile(startTime: Calendar, endTime:Calendar) :List<RecordFile>
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
startTime	Calendar	Req.	Start time
stopTime	Calendar	Req.	End time

Return Value

Return the video list of ***HPNetRecordFile*** type.

Example

Sample Code of Searching Video Segments

```
val stopTime = Calendar.getInstance()
val startTime = Calendar.getInstance()
startTime.timeInMillis = stopTime.timeInMillis – 24*3600*1000
try{
   val recordFiles = mPlayer.searchRecordFile(startTime,stopTime)
}catch(e:HPNetSDKException){
   //An exception occurred while searching.
}
```

3.4.3 startPlayback

Start the playback.

API Definition

```
- fun startPlayback(startTime: Calendar, stopTime:Calendar, safeKey:String?)
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
startTime	Calendar	Req.	Start time
stopTime	Calendar	Req.	End time
safeKey	String	Opt.	Video encryption verification code

Return Value

No value is returned. The playback result will be notified via the callback ***PlaybackCallback*** .

Example

Sample Code of Starting Playback

```
val stopTime = Calendar.getInstance()
val startTime = Calendar.getInstance()
startTime.timeInMillis = stopTime.timeInMillis – 24*3600*1000
mPlayer.startPlayback(startTime,stopTime, "123456")
```

3.4.4 pausePlayback

Pause the playback.

API Definition

```
- fun pausePlayback()
```

Return Value

No value is returned. The result of pausing the playback will be notified via the callback ***PlaybackCallback*** .

Example

Sample Code of Pausing the Playback

```
mPlayer. pausePlayback()
```

3.4.5 resumePlayback

Resume the playback.

API Definition

```
- fun resumePlayback()
```

Return Value

No value is returned. The result of resuming the playback will be notified via the callback ***PlaybackCallback*** .

Example

Sample Code of Resuming the Playback

```
mPlayer. resumePlayback()
```

3.4.6 openPlaySound

Turn on the audio during the live view or playback.

API Definition

```
- fun openPlaySound() :Boolean
```

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example
Sample Code of Turning On the Audio

```
val result = mPlayer.openPlaySound()
```

3.4.7 closePlaySound

Turn off the audio during the live view or playback.

API Definition

```
- fun closePlaySound():Boolean
```

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example
Sample Code of Turning Off the Audio

```
val result = mPlayer.closePlaySound()
```

3.4.8 stopPlay

Stop the live view or playback.

API Definition

```
- fun stopPlay() :Boolean
```

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example
Sample Code of Stopping the Live View or Playback

```
val result = mPlayer.stopPlay()
```

3.4.9 capturePicture

Capture pictures during the live view or playback.

API Definition

```
- fun capturePicture(filePath: String): Boolean
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
filePath	String	Req.	Path for saving files

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example
Sample Code of Capturing Pictures

```
val result = mPalyer.capturePicture(filePath)
```

3.4.10 startLocalRecord

Start recording during the live view or playback.

API Definition

```
- fun startLocalRecord(): Boolean
```

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example
Sample Code of Starting Recording

```
val result = mPlayer.startLocalRecord()
```

Remarks

- For the video storage path configuration of live view, refer to ***PreviewCallback*** .
- For the video storage path configuration of playback, refer to ***PlaybackCallback*** .

3.4.11 stopLocalRecord

Stop recording during the live view or playback.

API Definition

```
- fun stopLocalRecord(): Boolean
```

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example
Sample Code of Stopping Recording

```
val result = mPlayer.stopLocalRecord()
```

3.4.12 setPlaybackSpeed

Set the playback speed.

API Definition

```
- fun setPlaybackSpeed(@Type.PlaybackSpeed speed:Int)
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
speed	Int	Req.	The playback speed: PLAYBACK_SPEED_1 (normal forward), PLAYBACK_SPEED_2 (2x fast forward), PLAYBACK_SPEED_1_2 (1/2 slow forward), PLAYBACK_SPEED_4 (4x fast forward), PLAYBACK_SPEED_1_4 (1/4 slow forward), PLAYBACK_SPEED_8 (8x fast forward), PLAYBACK_SPEED_1_8 (1/8 slow forward).

Return Value

No value is returned. The result of setting the playback speed will be notified via the callback ***PlaybackCallback*** .

Example

Sample Code of Setting the Playback Speed

```
mPalyer.setPlaybackSpeed (Type.PLAYBACK_SPEED_1)
```

3.4.13 electricZoom

Zoom in or out during the live view and playback.

API Definition

```
- fun electricZoom(originalRect: RectF, targetRect: RectF): Boolean
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
originalRect	RectF	Req.	Original area
targetRect	RectF	Req.	Target area

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example

Sample Code of Digital Zoom

```
val result = mPlayer.electricZoom(originalRect, targetRect)
```

3.4.14 closeEletricZoom

Disable the digital zoom during the live view or playback.

API Definition

```
- fun closeElectricZoom(): Boolean
```

Return Value

Return *true* for success, and return *false* for failure.
If *false* is returned, you can call ***getLastError*** to get the error code for reference.

Example

Sample Code of Disabling Digital Zoom

```
val result = mPlayer.closeElectricZoom()
```

3.4.15 getCapacitySupport

Get the capability set. Currently, device supports getting the capability of speed playing only.

API Definition

```
@WorkerThread
- fun getCapacitySupport(capacityType: SupportItem): Boolean
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
capacityType	SupportItem	Req.	Capability type: PLAYBACK_SPEED (whether the device supports speed playing).

Return Value

The capability is supported when the API is successfully called and the return value is *true*, and is not supported when the return value is *false*.

Example

Sample Code of Getting the Capability Set

```
val result = mPlayer.getCapacitySupport(SupportItem.PLAYBACK_SPEED)
```

3.4.16 startDownloadFile

Start downloading a file.

API Definition

```
- fun startDownloadFile(
      devSerial: String,
      channelNo: Int,
      safeKey: String,
      startTime: Calendar,
      stopTime: Calendar,
      saveFilePath: String,
      downloadFileResultCallback: DownloadFileResultCallback
  )
```

Parameters

Parameter	Data Type	Req. or Opt.	Description
devSerial	String	Req.	Device serial No.
channelNo	String	Req.	Channel No.
safeKey	String	Req.	Verification code.
startTime	Calendar	Req.	Start time.
stopTime	Calendar	Req.	End time.
saveFilePath	String	Req.	The path to save the file.
downloadFileResultCallback	DownloadFileResultCallback	Req.	Callback of downloading result.

Return Value

No value is returned. The result will be notified via the callback ***DownloadFileResultCallback*** .

Example

Sample Code of Starting Downloading File

```
DownloadFileManager.startDownloadFile(
  devSerial,
  channelNo,
  secretKey,
  startTime,
  endTime,
  filePath,
  object : DownloadFileResultCallback {
     override fun onSuccess(downloadFilePath: String) {

     }

     override fun onError(errorCode: String) {

     }
  })
```

3.4.17 stopDownloadFile

Stop downloading a file.

API Definition

```
- fun stopDownloadFile()
```

Example

Sample Code of Stopping Downloading File

```
DownloadFileManager.stopDownloadFile()
```

Appendix A. Callback Function

A.1 InitCallback

Callback function of the SDK initialization

Initialization Succeeded

```
fun onInitSuccess()
```

Initialization Failed

```
fun onInitFail(error:HPNetError)
```

Table A-1 Parameters

Parameter	Data Type	Description
error	HPNetError	Error codes. Refer to ***HPNetError*** for details.

A.2 PreviewCallback

Callback function of the live view

Callback Function of the Live View Message

```
fun didReceivedMessage(player: IPlayer, messageCode:Int)
```

Table A-2 Parameters

Parameter	Data Type	Description
player	IPlayer	Live view instance
messageCode	Int	Message codes

Callback Function of the Live View Error

```
fun didOperateFail(player: IPlayer, @Type.Operate operateType:Int, error: HPNetError)
```

Table A-3 Parameters

Parameter	Data Type	Description
player	IPlayer	Live view instance
operateType	Int	Operation types, which may be returned during the live view: Type.OP_PLAY (start live view), Type.OP_RECORD (record), Type.OP_VOICE_TALK (two-way audio), Type.OP_SNAPSHOT (capture pictures), Type.OP_CHANGE_STREAM (switch stream), Type.OP_OTHER (other operations).
error	HPNetError	Error codes. Refer to ***HPNetError*** for details.

Callback Function of the Video Storage Path for Live View

```
fun getRecordFilePath(player: IPlayer) :String
```

Table A-4 Parameters

Parameter	Data Type	Description
player	IPlayer	Live view instance

Table A-5 Return Value

Description	Data Type
Video storage path	String

A.3 PlaybackCallback

Callback function of the playback

Callback Function of the Playback Message

```
fun didReceivedMessage(player: IPlayer, messageCode:Int)
```

Table A-6 Parameters

Parameter	Data Type	Description
Player	IPlayer	Playback instance
messageCode	Int	Message codes

Callback Function of the Playback Error

```
fun didOperateFail(player: IPlayer, @Type.Operate operateType:Int, error: HPNetError)
```

Table A-7 Parameters

Parameter	Data Type	Description
player	IPlayer	Playback instance
operateType	Int	Operation types, which may be returned during the playback: Type.OP_PLAY(start playback), Type.OP_PAUSE (pause playback), Type.OP_RESUME (resume playback), Type.OP_RECORD (record), Type.OP_SNAPSHOT (capture pictures), Type.OP_CHANGE_STREAMT (switch stream), Type.OP_ CHANGE_SPEED (change playback speed), Type.OP_OTHER (other operations)
error	HPNetError	Error codes. Refer to ***HPNetError*** for details.

Callback Function of the Playback Video Storage Path

```
fun getRecordFilePath(player: IPlayer) :String
```

Table A-8 Parameters

Parameter	Data Type	Description
Player	IPlayer	Playback instance

Table A-9 Return Value

Description	Data Type
Video storage path	String

A.4 IntercomCallback

Callback function of the two-way audio.

Callback Function of the Two-Way Audio Message

```
fun didReceivedMessage(intercom: IIntercom, msgCode:Int)
```

Table A-10 Parameters

Parameter	Data Type	Description
intercom	IIntercom	Two-way audio instance.
msgCode	Int	Message codes.

Callback Function of the Two-Way Audio Error

```
fun didReceiveError(intercom: IIntercom, error: HPNetError)
```

Table A-11 Parameters

Parameter	Data Type	Description
intercom	IIntercom	Two-way audio instance.
error	***HPNetError***	Error codes.

A.5 DownloadFileResultCallback

Callback function of the results of downloading files.

Callback Function of the Results of Downloading Files

```
fun onSuccess(downloadFilePath: String)
```

Table A-12 Parameters

Parameter	Data Type	Description
saveFilePath	String	The path of saving files.

Callback Function of Download Error

```
fun onError(errorCode: String)
```

Table A-13 Parameters

Parameter	Data Type	Description
errorCode	String	Error codes. Refer to ***HPNetSDK Error Code*** for details.

Appendix B. Data Structure

B.1 HPNetError

Error code instance

Property	Data Type	Description
code	String	Error code
message	String	Error description

B.2 HPNetPlayerParam

Player parameters

Property	Data Type	Description
devSerial	String	Device serial number
cannelNo	Int	The channel number
playWnd	SurfaceView	Player view

B.3 HPNetIntercomParam

Player parameters.

Property	Data Type	Description
devSerial	String	Device serial number
cannelNo	Int	The channel number

B.4 HPNetRecordFile

Video segment information

Property	Data Type	Description
startTime	Calendar	Start time of the video segment
endTime	Calendar	End time of the video segment
type	Int	Type of the video segment: 0 (scheduled recording), 1 (event recording)

B.5 HPNetSDKException

Error information

Property	Data Type	Description
code	String	Error code
message	String	Error description

Appendix C. HPNetSDK Status Code

Status Code	Description
101	Starting live view succeeded.
102	Switching stream succeeded.
103	Playing stopped automatically when the time limit of continuous live view has reached.
104	Playing stopped automatically when the token information changed during the live view.
105	The same channel of a device has been used.
201	Starting playback succeeded.
202	Playback ended.
203	Playback paused.
204	Playback resumed.
206	Playing stopped automatically when the time limit of continuous playback has reached.
207	Playing stopped automatically when the token information changed during the playback.
208	The same channel of a device has been used.
301	Starting two-way audio succeeded.
302	Two-way audio ended.
303	Playing stopped automatically when the time limit of continuous two-way audio has reached.
304	Playing stopped automatically when the token information changed during the two-way audio.

Appendix D. HPNetSDK Error Code

Error Code	Description
HPNETSDK1-100001	Protocol parsing error.
HPNETSDK1-100005	Network request timed out.
HPNETSDK1-100007	Network exception.
HPNETSDK1-100015	The player is not playing.
HPNETSDK1-100016	The SDK has not been initialized.
HPNETSDK1-100017	Duration of video to be downloaded exceeds the limit (5 minutes).
HPNETSDK1-100018	Incorrect parameter.
HPNETSDK4-400035	The verification code is required.
HPNETSDK4-400036 HPNETSDK4-320021 HPNETSDK4-120010	Verification code error.
HPNETSDK4-10070 HPNETSDK4-250000 HPNETSDK4-110031	No permission.
HPNETSDK4-120001	The channel does not exist.
HPNETSDK4-120002 HPNETSDK4-120007 HPNETSDK4-120023 HPNETSDK4-395404 HPNETSDK4-400901	The device is offline.
HPNETSDK4-120008 HPNETSDK4-390001 HPNETSDK4-395405	Device response timed out.
HPNETSDK4-320015 HPNETSDK4-390027 HPNETSDK4-390037 HPNETSDK4-390034	Streaming timed out.

Error Code	Description
HPNETSDK4-391009 HPNETSDK4-391209 HPNETSDK4-395411 HPNETSDK4-396520 HPNETSDK4-400034	
HPNETSDK4-400031 HPNETSDK4-400032	Network exception.
HPNETSDK4-395454	Streaming interrupted.
HPNETSDK4-395416 HPNETSDK4-395546 HPNETSDK4-395547	Device connection limit reached.
HPNETSDK4-000000 HPNETSDK4-110001	Parameter error.
HPNETSDK4-410030	Invalid device serial No.
HPNETSDK4-320027	Encapsulation failed.
HPNETSDK4-400097	Starting downloading failed.
HPNETSDK4-400098	Stopping downloading failed.
HPNETSDK4-400099	Unknown error.
HPNETSDK4-510000	Two-way audio via channel not supported by current device.


See Far, Go Further